

逐步构建过程

$X_s = XQ_s + E_{1,s}$ 找出选择矩阵 Q_s

$$\min_{Q_s} \sum_{s=1}^S \|Q_s\|_F^2 + \sum_{s=1}^S \|E_{1,s}\|_{2,1}$$

$$\text{s.t. } X_s = XQ_s + E_{1,s}$$

其中：

$X = [X_1, X_2, \dots, X_S] \in \mathbb{R}^{D \times N}$,是多模态数据, D为原始数据的特征数, N为所有模态数据的样本总数 $N = \sum_{s=1}^S n_s$

$X_s \in \mathbb{R}^{D \times n_s}$,是每个模态的数据

$P_s \in \mathbb{R}^{d \times D}$,d为降维之后特征数

$Q_s \in \mathbb{R}^{N \times n_s}$, (什么功能的矩阵?) n_s 为单个模态数据的样本数。

$Z_s \in \mathbb{R}^{n_s \times n_s}$

$E_s \in \mathbb{R}^{D \times n_s}$

加入低秩表示投影和HSIC约束

$$\min_{P, Z, Q, E_1, E_2} \sum_{s=1}^S \|Z_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_{1,s}\|_{2,1} + \delta \sum_{s=1}^S \|E_{2,s}\|_{2,1}$$

$$+ \kappa \sum_{s=1}^S \|P_s X Q_s - P_s X Q_s Z_s\|_F^2 + \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t)$$

$$\text{s.t. } X_s = XQ_s + E_{1,s}, XQ_s = XQ_s Z_s + E_{2,s}, P_s^T P_s = I$$

引入辅助变量J,Z=J

$$\min_{P, Z, Q, E_1, E_2, J} \sum_{s=1}^S \|Z_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_{1,s}\|_{2,1} + \delta \sum_{s=1}^S \|E_{2,s}\|_{2,1}$$

$$+ \kappa \sum_{s=1}^S \|P_s X Q_s - P_s X Q_s Z_s\|_F^2 + \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t)$$

$$\text{s.t. } X_s = XQ_s + E_{1,s}, XQ_s = XQ_s Z_s + E_{2,s}, Z_s = J_s, P_s^T P_s = I$$

构建增广拉格朗日函数

$$\begin{aligned}
& \mathcal{L}(P, Z, Q, E_1, E_2, J, Y_1, Y_2, Y_3) \\
&= \sum_{s=1}^S \|J_s\|_* + \beta \sum_{s=1}^S \|Q_s\|_F^2 + \gamma \sum_{s=1}^S \|E_{1,s}\|_{2,1} + \delta \sum_{s=1}^S \|E_{2,s}\|_{2,1} \\
&+ \kappa \sum_{s=1}^S \|P_s X Q_s - P_s X Q_s Z_s\|_F^2 + \lambda \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\
&+ \frac{\mu}{2} \|X_s - X Q_s - E_{1,s} + \frac{Y_1}{\mu}\| \\
&+ \frac{\mu}{2} \|X Q_s - X Q_s Z_s - E_{2,s} + \frac{Y_2}{\mu}\| \\
&+ \frac{\mu}{2} \|Z_s - J_s + \frac{Y_3}{\mu}\|
\end{aligned}$$

HSIC约束选择内积核函数 $K_s = Z_s^T Z_s$

$$\sum_{t=1, t \neq s}^S \text{HSIC}(Z_s, Z_t) = \sum_{t=1, t \neq s}^S \text{Tr}(H K_s H K_t) = \sum_{t=1, t \neq s}^S \text{Tr}(Z_s H K_t H Z_s^T) = \text{Tr}(Z_s K Z_s^T)$$

其中: $H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^T$ 是中心化矩阵, $K_t = Z_t^T Z_t$, $K = \sum_{t=1, t \neq s}^S H K_t H$
每次固定一个工况, 拉格朗日函数可以重写为:

$$\begin{aligned}
& \mathcal{L}(P, Z, Q, E_1, E_2, J, Y_1, Y_2, Y_3) \\
&= \|J_s\|_* + \beta \|Q_s\|_F^2 + \gamma \|E_{1,s}\|_{2,1} + \delta \|E_{2,s}\|_{2,1} \\
&+ \kappa \|P_s X Q_s - P_s X Q_s Z_s\|_F^2 + \lambda \text{Tr}(Z_s K Z_s^T) \\
&+ \frac{\mu}{2} \|X_s - X Q_s - E_{1,s} + \frac{Y_1}{\mu}\| \\
&+ \frac{\mu}{2} \|X Q_s - X Q_s Z_s - E_{2,s} + \frac{Y_2}{\mu}\| \\
&+ \frac{\mu}{2} \|Z_s - J_s + \frac{Y_3}{\mu}\|
\end{aligned}$$

更新

1. J ...

2. Z

```
kappa*tr((P*X*Q-P*X*Q*Z)*(P*X*Q-P*X*Q*Z)')  
+mu/2*tr((X*Q-X*Q*Z-E2+Y2/mu)*(X*Q-X*Q*Z-E2+Y2/mu)')  
+mu/2*tr((Z-J+Y3/mu)*(Z-J+Y3/mu)')  
+lambda*tr(Z*K*Z')
```

function:

$$\begin{aligned} f = & kappatr((PXQ - PXQZ)(PXQ - PXQZ)^\top) \\ & + mu/2tr((XQ - XQZ - E2 + Y2/mu)(XQ - XQZ - E2 + Y2/mu)^\top) \\ & + mu/2tr((Z - J + Y3/mu)(Z - J + Y3/mu)^\top) \\ & + lambdatr(ZKZ^\top) \end{aligned}$$

gradient:

$$\begin{aligned} \frac{\partial f}{\partial Z} = & mu(Z - J + 1/muY3) - 2kappa(PXQ)^\top(PXQ - PXQZ) \\ & - mu(XQ)^\top(XQ - XQZ - E2 + 1/muY2) \\ & + lambdaZK^\top + lambdaZK \end{aligned}$$

let =0

$$\begin{aligned} Z_{\text{set}\{i\}} = & \left(\mu + 2\kappa (P_{\text{set}\{i\}} XQ_{\text{set}\{i\}})^\top P_{\text{set}\{i\}} XQ_{\text{set}\{i\}} - \mu (XQ_{\text{set}\{i\}})^\top XQ_{\text{set}\{i\}} \right. \\ & \left. + \lambda (K^\top + K) \right)^{-1} \\ & \left(\mu J_{\text{set}\{i\}} - Y_{3_{\text{set}\{i\}}} + 2\kappa (P_{\text{set}\{i\}} XQ_{\text{set}\{i\}})^\top P_{\text{set}\{i\}} XQ_{\text{set}\{i\}} \right. \\ & \left. - \mu (XQ_{\text{set}\{i\}})^\top XQ_{\text{set}\{i\}} + \mu (XQ_{\text{set}\{i\}})^\top E_{2_{\text{set}\{i\}}} - Y_{2_{\text{set}\{i\}}} \right) \end{aligned}$$

3. P ...

4. Q

$$\min_Q \beta \|Q_s\|_F^2 + \kappa \|P_s X Q_s - P_s X Q_s Z_s\|_F^2 + \frac{\mu}{2} \|X_s - X Q_s - E_{1,s} + \frac{Y_1}{\mu}\|_F^2 \\ + \frac{\mu}{2} \|X Q_s - X Q_s Z_s - E_{2,s} + \frac{Y_2}{\mu}\|_F^2$$

function:

$$f = \text{betatr}(Q Q^\top) + \text{kappatr}((P X Q - P X Q Z)(P X Q - P X Q Z)^\top) \\ + \mu/2 \text{tr}((X Q - X Q Z - E2 + Y2/\mu)(X Q - X Q Z - E2 + Y2/\mu)^\top) \\ + \mu/2 \text{tr}((X_s - X Q - E1 + Y1/\mu)(X_s - X Q - E1 + Y1/\mu)^\top)$$

gradient:

$$\frac{\partial f}{\partial Q} = 2\text{beta}Q \\ + 2\text{kappa}(P X)^\top (P X Q - P X Q Z) - 2\text{kappa}(P X)^\top (P X Q - P X Q Z) Z^\top \\ + \mu X^\top (X Q - X Q Z - E2 + 1/\mu Y2) - \mu X^\top (X Q - X Q Z - E2 + 1/\mu Y2) Z^\top \\ - \mu X^\top (X_s - X Q - E1 + 1/\mu Y1)$$

5. E1

6. E2